



DigComp 3.0:

DigComp 3.0 – New Digital Competences Including AI & IP

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What is DigComp 3.0?

- ▶ A holistic, structured description of what is needed to be **digitally competent**
- ▶ Common **language** and shared **understanding**
- ▶ From **basic** to **highly advanced**
- ▶ Developed by the JRC in collaboration with DG EMPL, using **scientific** and **consultative** approaches



Who is DigComp 3.0 for, and how is it used?



- ▶ **Children and adults**
- ▶ **Employment, education, and training**
- ▶ A **transversal, generic** framework: a starting point to be adapted for a particular purpose





Definition of *digital competence*

*...the confident, **critical** and **responsible** use of, and engagement with, digital technologies for learning, at work, and for participation in society*

...as defined in the 2018 Council Recommendation for key Competencies in lifelong learning.



Why is DigComp 3.0 needed?



How was it developed?

- ▶ Supports EC priorities and policies
- ▶ Enables coherence in policymaking across the EU (and beyond)
- ▶ Based on scientific evidence and methods
- ▶ Is a large collaboration (JRC and +300 experts)

**Already
downloaded
21,000+ times**





People at the centre



Solidarity and inclusion



Freedom of choice



Participation



Safety and security



Sustainability

Grounded in the European declaration on digital rights and principles

Built on priority themes identified through literature, policy and expert consultation

IP is implicated in *AI* and in *rights, choice and responsibilities*



Content Themes

Artificial Intelligence (including generative AI) competence

Cybersecurity competence

Digital rights, choice and responsibilities

Wellbeing in digital environments

Competence to tackle misinformation and disinformation



Application Themes

Digital competence as a key part of lifelong learning

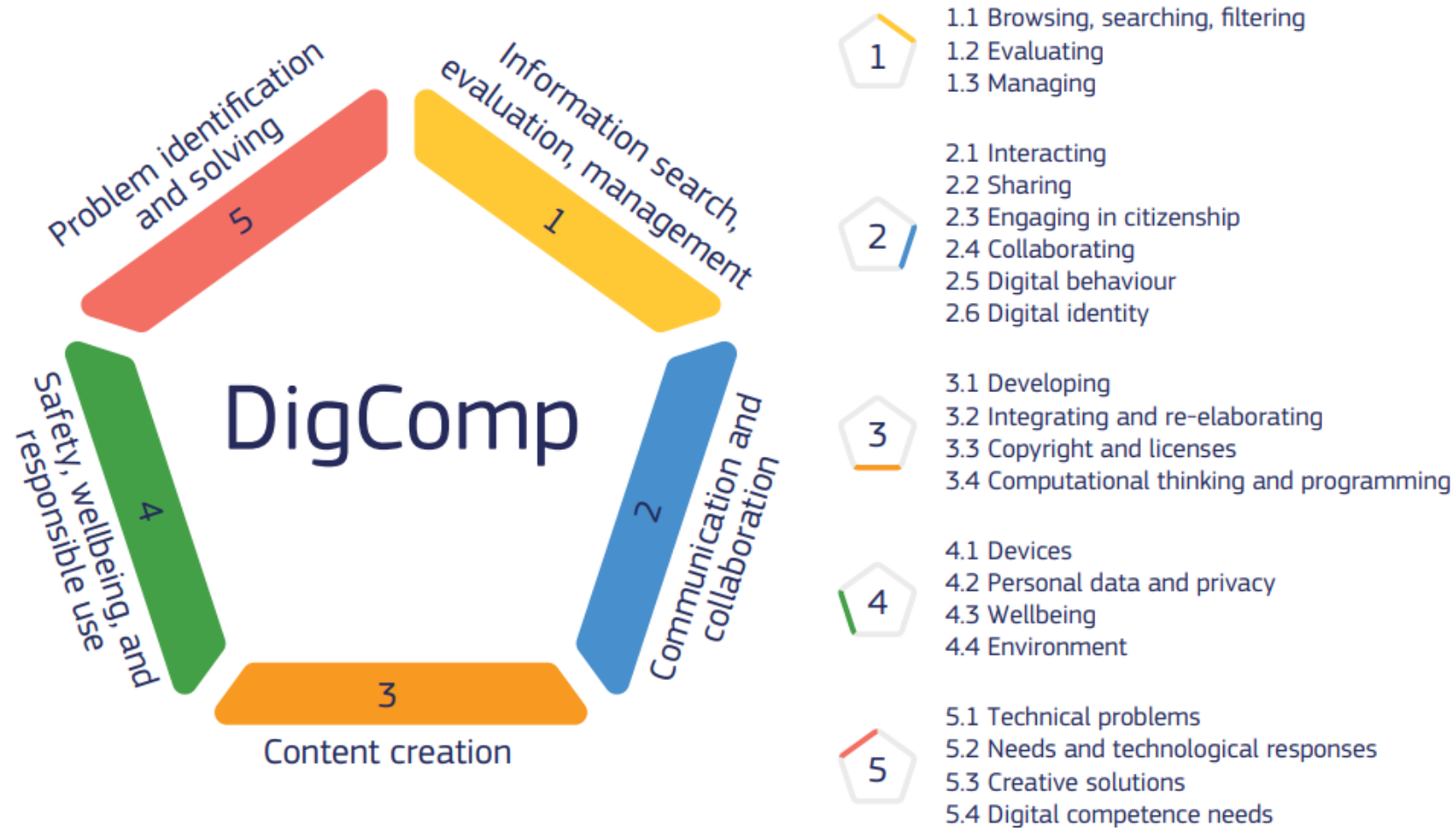
Recognition of pre-requisites for acquiring basic-level digital competence

Recognition of differences in digital competence needs across individuals and over time

Need for flexible, agile applications of the framework



DigComp 3.0 consists of **5 competence areas**, **21 competences**, and **four proficiency levels**...



Source: JRC own elaboration.

‘Horizontal’ or
content element

- At **Basic** level, individuals remember and implement simple tasks with guidance, as needed.
- At **Intermediate** level, they identify and implement well-defined tasks and solve well-defined problems autonomously.
- At **Advanced level**, individuals assess and apply solutions to a variety of complex tasks autonomously and adapt to a variety of contexts to evaluate and execute tasks appropriately, guiding others if and as required.
- At **Highly advanced level**, they assess, evaluate and resolve highly complex or specialised problems to create new solutions or adapt existing ones, leading and guiding others if and as required.

Source: JRC, own elaboration.

‘Vertical’ or competence acquisition element

Based on task complexity, cognitive demand and level of autonomy

Relevant sections in DigComp 3.0:

Section 2.1 – overview, page 16

Section 2.3 – competence areas and competences, page 17

Section 2.4 – proficiency levels, page 21

Section 3 – ‘integrated’ view of the framework, pages 29-49





New elements

Natural tension between continuity and change

DigComp 3.0 maintains the same overall structure, while having:

- ▶ Updates to the **wording** of competence areas and competences
- ▶ Revised **proficiency levels** including competence statements
- ▶ New **learning outcomes**
- ▶ Transversal integration of **AI competence**
- ▶ Extended **glossary**
- ▶ Multiple **formats**

Relevant sections in DigComp 3.0:

Section 1.5 – what is new in DigComp 3.0, page 12

Annex 1 – comparison of DigComp 2.2 and DigComp 3.0, pages 77-83





Learning outcomes

What an individual **knows or is able to** do after completing any learning process

- ▶ +500 learning outcomes
- ▶ Most detailed view of the framework
- ▶ Intended outcomes, not performance indicators

Relevant sections in DigComp 3.0:

Section 2.5 – learning outcomes (intro), page 23

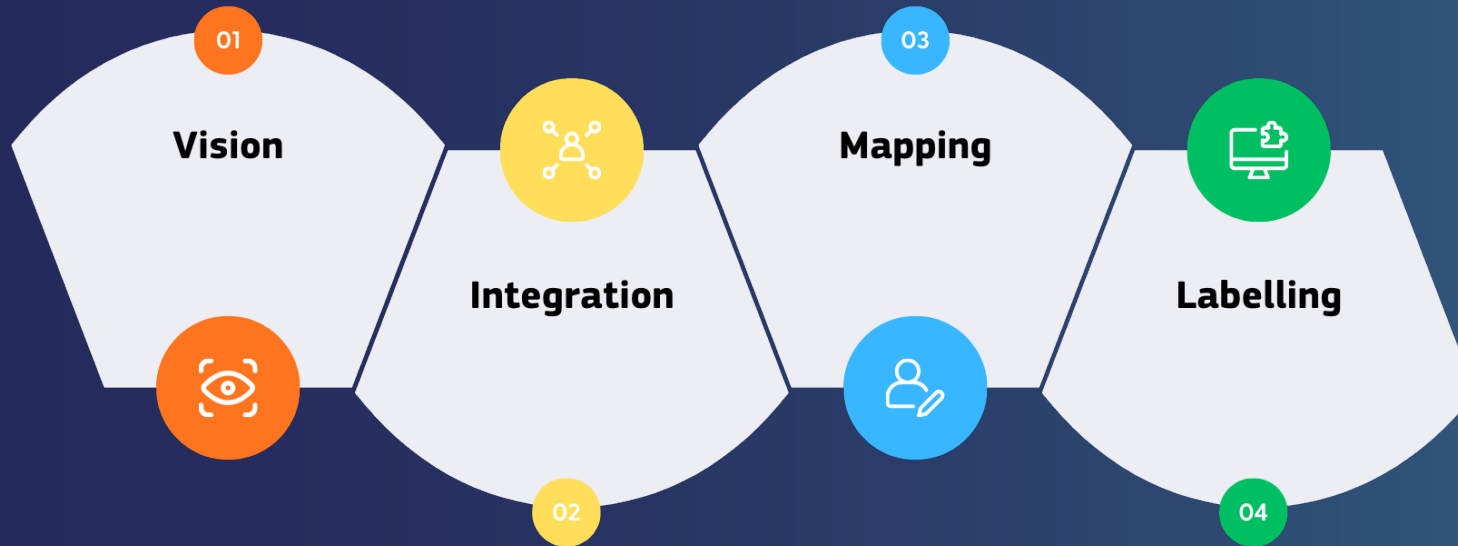
Annex 2 – learning outcomes (detail)

- > features, pages 84-85
- > practical considerations, page 86
- > full set of learning outcomes, pages 87-111





Integration of AI



AI is one digital technology among a range of digital technologies

Integration is **transversal**, appearing in all 21 competences

Approach is **holistic**, covering conceptual **understanding**, purposeful **use**, and ethical and societal **implications**

AI labelling:

AI-explicit [AI-E]

AI-implicit [AI-I]

Neither AI explicit nor implicit





DigComp 3.0: Links between elements



Focus on AI and IP (competence 3.3)



In DigComp 3.0 we extended competence 3.3, *copyright and licences*, to integrate **ethical** considerations and **AI**.

The competence description is more **action-oriented**.

To understand how copyright and licences, as well as associated legal and ethical issues, apply to digital content, and how to correctly apply them.



Focus on AI and IP (competence 3.3)



Basic

- > Recognise the **concepts** of **copyright** and **licence** in digital contexts, that individuals may need **permission** to use content found online, and that an individual's digital content is **automatically copyrighted**.
- > Identify digital content that can be used **free** of charge.
- > Recognise that **different kinds** of copyright and licences apply to digital content, **including AI-generated content**, and that these determine how content can be used and shared.
- > Recognise that **AI-generated content should be labelled** as such to help others understand its origin and possibilities for further use.
- > **Use and share** digital content in compliance with **basic legal and ethical guidelines**.

Intermediate

- > Prioritise a cautious approach (**check before using or sharing**), acknowledging the **complex** nature of digital copyright and licencing.
- > **Distinguish** between **copyright**, **trademark**, **design** and **patent** in digital contexts.
- > Identify **common types and purposes** of licences, including **Creative Commons**, and examples of **where copyright does and does not apply**.
- > Describe **legal**, **ethical** and **commercial** consequences of intellectual property violations, including **piracy** and **plagiarism**.
- > Identify examples of **legal** and **ethical** challenges relating to copyright in the **training of AI models**.
- > **Apply legal and ethical guidelines appropriately** when using and sharing digital content.

Focus on AI and IP (competence 3.3)



Advanced

- > **Prioritise** a consideration of **ethical and legal aspects**, such as **transparency** and **copyright**, when using and sharing digital content.
- > Describe **key features of current legislation** in relation to digital copyright and licences.
- > Identify **differences** in how **ethical** and **copyright** issues apply to **training** data for AI systems and AI-generated content (**output**).
- > **Assess and correctly apply** legal and ethical guidelines for using and sharing of digital content in **complex contexts**, including **software** licensing models and licence **renewal** requirements.
- > **Assist others** to use and share digital content in compliance with legal and ethical guidelines.

Highly advanced

- > **Stay informed about developments** in copyright and licensing regulations in digital contexts.
- > **Promote and support awareness and understanding** of legal and ethical copyright and licensing practices.
- > **Apply advanced knowledge** of intellectual property rights, copyright and licensing concepts in digital contexts **to inform decision-making**.
- > **Lead or contribute to policies or guidelines** on copyright and licensing in digital contexts.

Focus on AI and IP: Elsewhere



Competence 2.2 Sharing through digital technologies: acknowledge importance of ethical and responsible sharing of information and content.

Competence 3.1: Developing digital content: recognise that while AI systems can generate content, humans are essential to ensure ethical, responsible, and context-appropriate outputs.

Competence 3.2: Integrating and re-elaborating digital content: acknowledge importance of ethical and transparent practices when re-using or elaborating digital content



Conclusions



- ▶ DigComp 3.0 aims to be a clear, relevant and flexible tool to describe digital competence
- ▶ Given its generic and transversal nature, adaptations are likely to be needed – various formats are available
- ▶ The update reflects recent implications of AI and IP, not only in competence 3.3 but elsewhere, with more action orientation and focus on ethical as well as legal aspects





DigComp 3.0 resources

**Re-use
guidelines**

**PDF and editable
formats**

**Data
formats**

**Info
leaflet**

Errata



https://joint-research-centre.ec.europa.eu/projects-and-activities/education-and-training/digital-transformation-education/digital-competence-framework-digcomp/digcomp-30-resources_en

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Thank you!

